

# Impact of the different levels of optimization for project 3

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O0-

For this executable, there are no optimizations made to the file. It runs exactly how I typed it originally. The impact of this is that it is the slowest out of the other optimizations or O1 and O3.

O1-

For this executable, there are only basic optimizations provided in the code. For example, this optimization level will reduce the complexity of the for loops I made in this program. It also overall reduces the amount of redundant code that is left in my program and creates more simple and cleaner loops. The impact of this is that it is faster than no optimization and it does not aggressively change the code in a meaningful way.

O3-

For this executable, there are more aggressive optimizations that are being implemented. For example, just like in the O1 optimization, it reduces the loop complexity but this time more aggressively. It also compacts the logic of the code making it way smaller than the unoptimized code and smaller than the O1 optimized code. Overall, this optimization's impact is more severe and causes the code to be run faster though in a more rough and aggressive optimized way.